Chapter 3

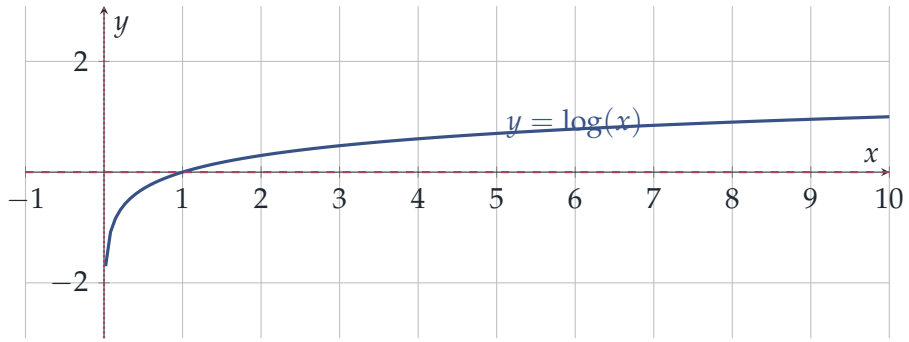
Common logarithm function

Skills: converting between exponential and logarithmic form, solving, modelling

3.1 Reference notes

Key points

For every $b > 0$, $\log(b)$ is the unique real number x such that $10^x = b$. Hence $\log(10^x) = x$ and, for $b > 0$, $10^{\log(b)} = b$. The function log is defined on $]0, +\infty[$ and it is strictly increasing.



For $a > 0$, $b > 0$ and every natural number n:

$$\log(ab) = \log(a) + \log(b), \qquad \log\left(\frac{a}{b}\right) = \log(a) - \log(b), \qquad \log(a^n) = n\log(a).$$

Common mistake

$\log(a + b)$ does not simplify. Before any logarithmic calculation, check that each argument is strictly positive.

Method

To solve $a^x = b$, with $a > 0$, $a \neq 1$ and $b > 0$, take the logarithm: $x\log(a) = \log(b)$, so $x = \dfrac{\log(b)}{\log(a)}$.

Worked example

Solving $1.08^x = 2$ amounts to writing $x = \dfrac{\log(2)}{\log(1.08)} \approx 9.01$. A value multiplied by 1.08 each year therefore doubles after a little more than nine years.

Since log is increasing, for $a, b > 0$, $a < b$ is equivalent to $\log(a) < \log(b)$. For $a > 1$, $a^x < b$ is equivalent to $x < \dfrac{\log b}{\log a}$; when $0 < a < 1$, the inequality sign reverses.

Session 15 — Definition of the common logarithm

Common logarithm function

17 Nov 2026 • B: 2 h – C: 1 h

Session objectives

- Link powers of 10 and logarithms.
- Calculate common values.
- Use the calculator.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 20 % of 80. .....
2. Give the domain of log. .....
3. Calculate $\log(10^4)$. .....
4. Solve $10^x = 0.01$. .....
5. Expand $\log(3b)$, $b > 0$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: link powers of 10 and logarithms and calculate common values.

Investigation

Complete the pairs $10^x = b$ and $x = \log(b)$ for $b = 0.001; 0.1; 1; 10; 1\,000$. Then use the calculator for $b = 7$.

Written record

For $b > 0$, $\log(b)$ is the unique solution of $10^x = b$. In particular $\log(1) = 0$, $\log(10) = 1$ and $\log(100) = 2$.

Worked example

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Practice – essential

1. Calculate $\log(10^2)$ and $\log(10^{-2})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $2^x = 22$ and give an approximate value to the nearest hundredth.

Textbook references

Further reading: Calao, course pp. 64–69; exercises p. 70, no. 1–5. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 70, exercise 2

Solve the following equations.

a. $5 \times 10^x = 3.375$

b. $3.2 + 2 \times 10^x = 4.5 \times 10^x$

c. $-17.3 + 10^x = 5 - 3 \times 10^x$

Homework

Redo practice exercise 2 with the following data: replace the initial value by 195 and the rate or common ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 16 — Variations and graph

Common logarithm function

19 Nov 2026 • B: 1 h – C: 2 h

Session objectives

- Know the domain.
- Use the fact that the function is increasing.
- Read the curve.

Fluency – 6 to 8 minutes, no calculator

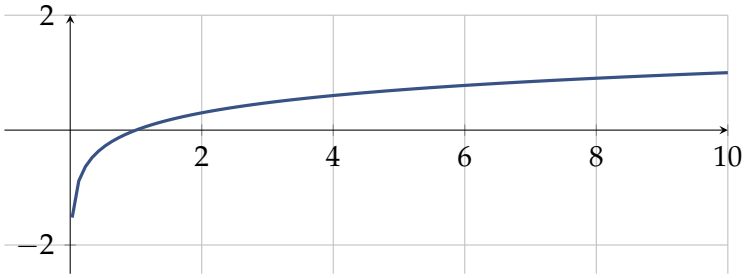
1. Write the multiplier for an increase of 25 %. .....
2. Calculate $\log(10^0)$. .....
3. Solve $10^x = 0.01$. .....
4. Expand $\log(3b)$, $b > 0$. .....
5. Give the domain of log. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: know the domain and use the fact that the function is increasing.

Investigation

Plot $y = 10^x$ and the line $y = x$, then obtain the curve of the logarithm by symmetry. Deduce its domain, its direction of variation and its asymptote.



Written record

The function log is defined on $]0, +\infty[$, is strictly increasing, and its curve has $x = 0$ as a vertical asymptote.

Worked example

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Practice – essential

1. Calculate $\log(10^3)$ and $\log(10^{-3})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $3^x = 23$ and give an approximate value to the nearest hundredth.

Textbook references

Further reading: Calao, course pp. 64–69; exercises p. 70, no. 6–10. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 70, exercise 7

Give the sign of the following numbers.

a. $\log(2.5)$ b. $\log(0.25)$ c. $\log\left(\dfrac{7}{10}\right)$

Homework

Redo practice exercise 3 with the following data: replace the initial value by 200 and the rate or common ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 17 — Algebraic properties

Common logarithm function

24 Nov 2026 • B: 2 h – C: 1 h

Session objectives

- Expand or condense logarithms.
- Check the conditions.
- Avoid false properties.

Fluency – 6 to 8 minutes, no calculator

1. Simplify $3x + 6 - x + 2$. .....
2. Solve $10^x = 0.01$. .....
3. Expand $\log(3b)$, $b > 0$. .....
4. Give the domain of log. .....
5. Calculate $\log(10^1)$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: expand or condense logarithms and check the conditions.

Investigation

Starting from $10^2 \times 10^3 = 10^5$, conjecture $\log(ab) = \log a + \log b$. Test numerically with $a = 2, b = 5$, then explain why a, b must be positive.

Written record

For $a, b > 0$, $\log(ab) = \log a + \log b$ and $\log\left(\frac{a}{b}\right) = \log a - \log b$. Moreover $\log(a^n) = n \log a$.

Worked example

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Practice – essential

1. Calculate $\log(10^4)$ and $\log(10^{-4})$.
2. Expand $\log\left(\frac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $4^x = 24$ and give an approximate value to the nearest hundredth.

Textbook references

Further reading: Calao, course pp. 66–69; exercises p. 72, no. 20–23. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 73, exercise 24

Write the following numbers in the form $\log(A)$, where A is a real number to be specified:

a. $\log(2) + \log(7) - \log(5)$

b. $\log(3) - 2\log(5)$

c. $\log(3) + \log(7)$

d. $3\log(7) - 7\log(3)$

e. $\log(12) - \log(4) + 2\log(3)$

Homework

Redo practice exercise 2 with the following data: replace the initial value by 205 and the rate or common ratio by 7. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 18 — Solving $10^x = b$

Common logarithm function

26 Nov 2026 • B: 1 h – C: 2 h

Session objectives

- Solve $10^x = b$.
- Bracket a solution.
- Interpret an order of magnitude.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 5 % of 140. .....
2. Expand $\log(3b)$, $b > 0$. .....
3. Give the domain of log. .....
4. Calculate $\log(10^2)$. .....
5. Solve $10^x = 0.01$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: solve $10^x = b$ and bracket a solution.

Investigation

A device measures intensities $b = 0.04; 3.2; 850$. For each value, solve $10^x = b$, give x to the nearest hundredth and check by calculating 10^x.

Written record

The equation $10^x = b$ has a solution only if $b > 0$; this solution is $x = \log(b)$.

Worked example

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Practice – essential

1. Calculate $\log(10^5)$ and $\log(10^{-5})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $5^x = 25$ and give an approximate value to the nearest hundredth.

Textbook references

Further reading: Calao, p. 73, no. 30–32. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 73, exercise 32

Harpagon placed a sum of 24 000 € with compound interest at an annual rate of 3.5 %. Some time later, he decides to close the account when the balance is 40 208.37 €. The aim of the exercise is to determine how many years the account existed.

1. Complete the unfinished algorithm so that it solves the problem.
2. Determine the number of years the account existed.
3. Translate the algorithm into Python and run it in a Python editor. Compare the result with that of question 2.

Homework

Redo practice exercise 3 with the following data: replace the initial value by 210 and the rate or common ratio by 2. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 19 — Solving $a^x = b$

Common logarithm function

1 Dec 2026 • B: 2 h – C: 1 h

Session objectives

- Solve $a^x = b$.
- Use a quotient of logarithms.
- Round sensibly.

Fluency – 6 to 8 minutes, no calculator

1. Write the multiplier for an increase of 10 %. .....
2. Give the domain of log. .....
3. Calculate $\log(10^3)$. .....
4. Solve $10^x = 0.01$. .....
5. Expand $\log(3b)$, $b > 0$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: solve $a^x = b$ and use a quotient of logarithms.

Investigation

The number of subscribers is multiplied by 1.15 each month. Determine after how many months it will have tripled. Set up the equation, solve it using logarithms and interpret the rounding.

Written record

For $a > 0$, $a \neq 1$ and $b > 0$, $a^x = b$ is equivalent to $x = \dfrac{\log(b)}{\log(a)}$.

Worked example

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Practice – essential

1. Calculate $\log(10^6)$ and $\log(10^{-6})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $6^x = 26$ and give an approximate value to the nearest hundredth.

Textbook references

Further reading: Calao, p. 74, no. 33–39. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 73, exercise 30

1. A first capital of 6 000 € is placed with compound interest at a rate of 9 %. After how many years will this capital have doubled? Tripled?
2. A capital of 9 000 € is placed on the same day at a rate of 6 %. After how many years will the value of the first capital have exceeded the second?

Homework

Redo practice exercise 2 with the following data: replace the initial value by 215 and the rate or common ratio by 3. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 20 — Equations $x^a = b$ and simple inequalities

Common logarithm function

3 Dec 2026 • B: 1 h – C: 2 h

Session objectives

- Solve equations where the variable is the base.
- Deal with an inequality.
- Take the domain into account.

Fluency – 6 to 8 minutes, no calculator

1. Simplify $3x + 9 - x + 2$. .....
2. Calculate $\log(10^4)$. .....
3. Solve $10^x = 0.01$. .....
4. Expand $\log(3b)$, $b > 0$. .....
5. Give the domain of log. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: solve equations where the variable is the base and deal with an inequality.

Investigation

Solve on $[0, +\infty[$: $x^3 = 40$, then $x^4 = 16$. Then solve $1.08^x < 1.5$ and explain the role of the function being increasing.

Written record

To solve $x^n = b$, take the domain and the parity of n into account. For an exponential inequality, the direction depends on whether the function is increasing or decreasing.

Worked example

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Practice – essential

1. Calculate $\log(10^2)$ and $\log(10^{-2})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $2^x = 22$ and give an approximate value to the nearest hundredth.

Textbook references

Further reading: Calao, p. 75, no. 40–42. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 74, exercise 36

A capital of 15 500 € is placed with compound interest for 4 and a half years. A second capital of 16 480 € is also placed with compound interest for 5 years and 3 months at an annual rate of 6 %. What must the rate of the first investment be so that the capitals at the end of the investments are equal?

Homework

Redo practice exercise 3 with the following data: replace the initial value by 220 and the rate or common ratio by 4. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Session 21 — Problems in context

Common logarithm function

8 Dec 2026 • B: 2 h – C: 1 h

Session objectives

- Model a situation.
- Choose an equation.
- Conclude in context.

Fluency – 6 to 8 minutes, no calculator

1. Calculate 20 % of 60. .....
2. Solve $10^x = 0.01$. .....
3. Expand $\log(3b)$, $b > 0$. .....
4. Give the domain of log. .....
5. Calculate $\log(10^3)$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: model a situation and choose an equation.

Investigation

The sound level is $L = 10\log\left(\frac{I}{I_0}\right)$. Calculate L when $I = 10^6 I_0$. Then determine the ratio $\frac{I}{I_0}$ for $L = 85$ dB.

Written record

In a problem, define the variable, set up the equation, solve, check the domain and conclude with the unit. A numerical solution without interpretation is incomplete.

Worked example

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Practice – essential

1. Calculate $\log(10^3)$ and $\log(10^{-3})$.
2. Expand $\log\left(\frac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $3^x = 23$ and give an approximate value to the nearest hundredth.

Textbook references

Further reading: Calao, p. 74, no. 38–39, then p. 75, no. 40–42. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 75, exercise 41

The dynamic range of an image is the ability of a digital image to reproduce the scales of light intensity of the real world. Dynamic range is often expressed in stops.
In digital imaging, the darkest pixel (black) corresponds to 0 and the lightest pixel (white) corresponds to 255. There are thus 256 light intensities, which in computing units amounts to 8 bits of precision ($256 = 2^8$). The number of stops is: $\dfrac{\log(256)}{\log(2)} = 8$.

1. For an underexposed digital photo, only 128 light intensities can be captured. How many stops does this correspond to?
2. A RAW file has a precision of 10 bits. What is the corresponding number of stops?
3. A standard video camera has an image dynamic range of 5.5 stops. How many different light intensities does it capture?
4. The latest-generation television screens have an image dynamic range of 16 stops. How many different light intensities can they display?
5. The number of light intensities in the real world is of the order of 100 000 000. How many stops does this correspond to?

Homework

Redo practice exercise 2 with the following data: replace the initial value by 225 and the rate or common ratio by 5. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Session 22 — Review and assessment

Common logarithm function

10 Dec 2026 • B: 1 h – C: 2 h

Session objectives

- Summarise the methods.
- Identify the conditions.
- Correct mistakes.

Fluency – 6 to 8 minutes, no calculator

1. Write the multiplier for an increase of 25 %. .....
2. Expand $\log(3b)$, $b > 0$. .....
3. Give the domain of log. .....
4. Calculate $\log(10^4)$. .....
5. Solve $10^x = 0.01$. .....

Reminder and prerequisites

Reread the last written record and be able to carry out the basic calculations needed today. Before the investigation, have students rephrase the key words: summarise the methods and identify the conditions.

Investigation

Make a map of methods: domain; properties; equations $10^x = b$; equations $a^x = b$; inequalities. Attach a condition and an example to each branch.

Written record

Before calculating a logarithm: check positivity. Before a comparison: identify the direction of variation. Before a conclusion: check that the result is plausible.

Worked example

Revisit the data from the investigation and write up the solution in order: definition of the objects, property used, exact or approximate calculation, then a concluding sentence. The model answer appears in the teacher edition.

Practice – essential

1. Calculate $\log(10^4)$ and $\log(10^{-4})$.
2. Expand $\log\left(\dfrac{100a^2}{b}\right)$ for $a, b > 0$.
3. Solve $4^x = 24$ and give an approximate value to the nearest hundredth.

Textbook references

Further reading: Calao, pp. 76–79. The exercise reproduced in this session is identified below by its page and number; the reference also lets you find the figures and digital resources of the textbook.

Calao textbook – p. 75, exercise 42

N is a nonzero natural number. The aim of the exercise is to find a method for determining the number of digits in its decimal representation.

1. $N = 10\,203$.
   a. Bound N between two consecutive powers of 10. Deduce a bound for $\log(N)$ between two consecutive integers.
   b. Deduce the value of $\log(N)$ rounded up to the nearest unit. Compare this result with the number of digits of N.
2. N has 23 digits.
   a. Bound N between two consecutive powers of 10. Deduce a bound for $\log(N)$ between two consecutive integers.
   b. Give the value of $\log(N)$ rounded up to the nearest unit. Which value do we recover?
3. From the previous questions, deduce a method for determining the number of digits of any nonzero natural number given in decimal form.
4. Use the previous method to determine the number of digits of the following integers: a. 7^{49}; b. 5658; c. $2\,019 \times 10^{20}$.
5. The largest known prime number to date is a Mersenne number written as $2^{82\,589\,933} - 1$. How many digits does it have?

Homework

Redo practice exercise 3 with the following data: replace the initial value by 230 and the rate or common ratio by 6. Write it up in four lines: data, formula, calculation, conclusion.

Exit ticket – 3 minutes

On a half-sheet: write the main rule of the session, work through a very short example and state your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.